

RESEARCH STUDY — FEBRUARY 2026

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:
US/Europe Divergence Trajectories 2024–2030

Integrated Geostrategic and Economic Analysis

Chapter IV Quarter

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77% global AI compute = USA **1.6×** energy cost EU/US
3.4:1 CACI ratio US/EU (Power Mode)

Paris — February 2026

7 chapters • 4 prospective scenarios • 3 geographic zones

Keywords: artificial intelligence, technology protectionism, semiconductors,
export controls, sovereign compute, AI geopolitics, France, United States,
China

Consequences for Africa

The analysis conducted in the preceding chapters covered Europe, South America, and Asia. However, Africa — identified in the General Conclusion as a necessary research extension — now constitutes a decisive geopolitical battleground. Representing 18% of the world's population but less than 1% of global data center capacity, the continent combines a structural compute deficit with energy potential (geothermal, solar, hydroelectric) and demographic dynamism that make it a major stake in the reshaping of the global technological order.¹ This supplementary chapter analyzes the specific consequences of US AI protectionism on Africa, distinguishing sub-regional dynamics and the differentiated effects of US-China rivalry.

6quat.1 Structural Position: The Continent of the “Compute Paradox”

6quat.1.1 An Infrastructure Gap Without Equivalent

Africa exhibits the most extreme compute asymmetry in the world. As of mid-2025, the continent has 223 data centers spread across 38 countries — less than 2% of the global total (11,800+). Installed capacity is estimated between 780 MW and 1,170 MW, equivalent to a single US hyperscale campus.² The distribution is itself heavily concentrated: South Africa (56 data centers), Kenya (19), and Nigeria (17) host 41% of the continental infrastructure. Egypt and Morocco dominate North Africa.

In terms of AI GPUs — the determining factor for frontier compute — the gap is even more stark. Only 5% of African AI innovators have access to advanced compute, while the unmet demand deficit is estimated at 7 million GPU hours over three years.³ This gap does not reflect an absence of demand but an *absence of supply*: the African compute paradox lies in the coexistence of growing latent demand and insufficient infrastructure to serve it. The WEF estimates that breaking this paradox could unlock \$1.2 to \$1.5 trillion in economic value by 2030.

Growth is nonetheless rapid. McKinsey (November 2025) projects African data center demand between 1.5 and 2.2 GW by 2030, with a compound annual growth rate of 14.5% to 24% depending on the segment. The African data center market, valued at \$1.94 billion in 2025, is expected to reach \$4.36 billion by 2031.⁴ The Nigerian market alone is valued at \$288 million in 2025 with a projection to \$1.09 billion by 2031 (CAGR 25%).

6quat.1.2 Tier Classification and Access to US Compute

Under the US export control architecture, virtually all African countries fall under **Tier 2** (restricted access with quantitative caps), or even **Tier 3** for sanctioned countries. No African country is in Tier 1 (unlimited access), reserved for the 20 closest US allies.

This classification has direct consequences: GPU caps impose a structural ceiling on frontier compute infrastructure accessible to African actors, even when funding is available. The cost of Nvidia GPUs on the secondary market is estimated between \$45,000 and \$60,000 per unit.⁵

Paradoxically, this US restriction opens a strategic space for Chinese suppliers. Huawei, present on the continent since the 1990s through its telecom equipment, controls approximately 70% of Africa's 4G backbone and leads 5G deployment in 30 markets. The company now offers integrated data center solutions (FusionModule2000, UPS5000-E) including Ascend AI accelerators, at significantly lower prices than Nvidia.⁶ The Tier 2 classification thus creates a *push factor* toward the Chinese technological ecosystem, reproducing in Africa the mechanism identified for South America (Chapter VI bis).

6quat.2 Emerging Hubs: Differentiated Mapping

6quat.2.1 South Africa: Primary Hub and Hyperscaler Anchor Point

South Africa holds 40.8% of the African data center market share in 2025. The South African AI data center market is growing at 42% annually (from \$70 million in 2025 to \$572 million by 2031). The three US hyperscalers have consolidated their presence: Microsoft (ZAR 5.4 billion, ~\$300 million by 2027, following ZAR 20.4 billion invested over three years), AWS (\$1.7 billion in the Cape region), Google (Johannesburg cloud region).⁷ The country benefits from a National AI Strategy, the Centre for AI Research (CAIR), and a relatively developed electrical grid, despite recurring load-shedding episodes. It is also the headquarters of Africa's first Cassava Technologies/Nvidia AI Factory (3,000 GPUs deployed mid-2025).

6quat.2.2 Nigeria: West Africa's Growth Engine

Nigeria is emerging as Africa's fastest-growing data center market. Lagos concentrates the bulk of investments: Equinix (\$100 million continental expansion plan), Africa Data Centres, Digital Realty, Rack Centre.⁸ The country benefits from its submarine connectivity (WACS, ACE, MainOne, Glo-1 cables) and a dynamic fintech ecosystem (13 deposit banks integrating AI chatbots by 2024). Microsoft is committed to training one million Nigerians in AI, with the country having identified \$15 billion in potential AI contribution to GDP by 2030. The National Centre for AI and Robotics (NCAIR) structures the research ecosystem.

6quat.2.3 Kenya: The "Silicon Savannah" and the Geothermal Advantage

Kenya stands out through the unique combination of a dynamic tech ecosystem ("Silicon Savannah") and a structural energy advantage: over 60% of its electricity comes from renewable sources, including geothermal from the Naivasha zone. This advantage attracted the \$1 billion investment from Microsoft and G42 (UAE) for a 100 MW geothermal data center in Olkaria, expandable to 1 GW.⁹ The government offers

tax incentives in special economic zones (10% corporate tax exemption for 10 years). The IXAfrica campus in Nairobi, in strategic partnership with Safaricom, already provides AI-ready infrastructure. The national AI strategy (March 2025) positions Kenya as a regional leader in AI research and innovation.

Kenya is also a testing ground for US-Kenya semiconductor partnerships: the USTDA partnered with Kenya Semiconductor Technologies Limited to establish a semiconductor manufacturing facility in Nairobi, an unprecedented initiative in sub-Saharan Africa.¹⁰

6quat.2.4 Morocco: The Europe-Africa Compute Node

Morocco occupies a uniquely strategic position: situated 15 km from Europe, connected via multiple submarine cables, it is positioning itself as an *AI compute export hub* rather than a domestic market. In June 2025, a consortium led by Naver (South Korea) with Nvidia announced a 500 MW data center project powered by renewable energy, with a first phase of 40 MW equipped with Blackwell GB200 GPUs. Morocco represents approximately 35% of the continental data center capacity (~140 MW).¹¹ The government's target is ambitious: 40 billion dirhams in advanced digital service revenues by 2030, primarily serving European and Middle Eastern clients. Iozera (Texas) had already announced a \$500 million investment for a 386 MW data center in Tetouan. The International AI Center affiliated with Mohammed VI Polytechnic University, designated a UNESCO Category II center in late 2023, anchors research.

6quat.2.5 Egypt: Regional Ambition and Government Strategy

Egypt has adopted one of the continent's most structured AI strategies, with an evaluation framework based on three criteria (technological feasibility, societal impact, return on investment). The country benefits from its geographical position as a connection point between Europe, Africa, and the Middle East via major internet backbones. Egypt is among the five target countries for the Cassava/Nvidia AI Factory (12,000 GPUs total over three to four years).¹²

6quat.2.6 Rwanda: Africa's "AI Laboratory"

Rwanda positions itself as "Africa's AI laboratory" (the official formulation of its National Policy on AI). The country hosted in April 2025 the first Global AI Summit on Africa in Kigali, bringing together participants from 40+ countries. Rwanda is the only African country with a formal national AI policy (distinct from a strategy). The Centre for the Fourth Industrial Revolution (C4IR Rwanda), in partnership with the WEF, experiments with AI solutions in health, agriculture, education, and public services. The Masakhane initiative, partly hosted in Rwanda, develops NLP models for African languages.¹³

Table 20 — Mapping of African AI Hubs

Hub	DCs (#)	Key Investment	Distinctive Asset	Dominant Player
South Africa	56	MS \$300M; AWS \$1.7B; Cassava/Nvidia 3,000 GPUs	Top African DC market (40.8%); CAIR; National AI Strategy	US Hyperscalers
Nigeria	17–20	Equinix \$100M; DC market \$288M (2025) → \$1.09B (2031)	220M population; fintech; Lagos submarine cables	US/Local mix
Kenya	19	MS+G42 \$1B (geothermal 100MW → 1GW); IXAfrica	60%+ geothermal; Silicon Savannah; SEZs	MS / G42 (UAE)
Morocco	~15	Naver/Nvidia 500MW; Iozera \$500M; Cassava AI Factory	Proximity to Europe (15 km); renewables; compute export	Korea / US
Egypt	~12	Cassava/Nvidia; Middle East/Africa hub	Mature AI strategy; crossroads geographic position	US/Gulf mix
Rwanda	3–5	C4IR; Kigali AI Summit 2025; Masakhane NLP	Formal AI policy; continental “AI lab”	Multilateral/WEF

6quat.3 US-China Rivalry in Africa: AI as the New Battleground

6quat.3.1 The US Offensive: Nvidia, Microsoft and the AI Factory Model

The most structuring US engagement is the Cassava Technologies-Nvidia partnership, announced in March 2025. This project constitutes one of the largest private technology infrastructure commitments in Africa: up to \$720 million to deploy 12,000 Nvidia GPUs across five countries (South Africa, Egypt, Nigeria, Kenya, Morocco) over three to four years. Nvidia subsequently made an equity investment in Cassava in October 2025.¹⁴ The model is AI-as-a-Service (AIaaS): making GPUs available for rent to African startups, universities, and governments, partially circumventing the prohibitive cost of direct acquisition.

In parallel, US hyperscalers are intensifying their presence: Microsoft (\$300 million in South Africa, \$1 billion in Kenya with G42), Google (\$37 million specifically for “AI for Africa,” within a global \$1 billion commitment to African tech). Microsoft has trained over 4 million young Africans in digital skills over five years and is committed to connecting 124 million people in Africa to the internet.¹⁵ The US AI Action Plan (July 2025) explicitly integrates Africa into its strategy for exporting the “full AI technology stack.”

6quat.3.2 Chinese Penetration: Huawei, DeepSeek, and Digital Belt and Road

Chinese presence in Africa long predates the AI competition. Huawei built much of the continent's telecom infrastructure under the Belt and Road Initiatives and dominates 70% of the African 4G backbone with operations in 30+ markets.¹⁶ This installed base confers a considerable structural advantage for the AI layer. Huawei is now deploying integrated AI data centers: in Ethiopia, the FusionModule2000 was installed at the Cooperative Bank of Oromia (\$24.7 billion in digital transactions), making Huawei the first to deploy operational AI infrastructure in sub-Saharan Africa, ahead of Schneider Electric, Vertiv, and Eaton still in pilot phase.

The Chinese chatbot DeepSeek is rapidly spreading in Kenya, where startups are using it for economic analysis and investment risk assessment, at a cost 94% lower than ChatGPT. Huawei has trained 120,000 Africans and is developing local supply chains.¹⁷ The 2024 FOCAC (Forum on China-Africa Cooperation) announced the development of a China-Africa digital cooperation center and 20 digital infrastructure and transformation projects. The parallel with physical Belt and Road is direct: Chinese AI infrastructure improves access but creates technological and political dependency.

6quat.3.3 The African “Double Bind”

Africa is caught in a structural *double bind*. On one hand, US protectionism (Tier 2 classification, 25% Section 232 tariffs) restricts access to Nvidia frontier compute. On the other, resorting to the Chinese alternative (Huawei Ascend, DeepSeek) exposes countries to surveillance risks, geopolitical dependency, and secondary US sanctions. DeepSeek user data is stored on servers accessible to the Chinese government. For African countries aspiring to maintain relationships with both blocs, every infrastructure choice is an alignment choice.¹⁸

Table 21 — US-China Competition in Africa: AI Engagement Mapping

Dimension	United States	China
Infrastructure	Cassava/Nvidia (\$720M, 12,000 GPUs); MS (\$300M SA, \$1B Kenya); AWS (\$1.7B SA); Google (\$1B global)	Huawei 70% 4G backbone; integrated DCs Ethiopia; 30 5G markets; FOCAC 20 digital projects
AI Models	ChatGPT (via US cloud); Nvidia AI stack; OpenAI/Penda Health (16% diagnostic error reduction Kenya)	DeepSeek (−94% cost vs ChatGPT); Alibaba Qwen; Huawei Pangu
Talent / Skilling	MS: 1M SA + 1M Nigeria + 1M Kenya in AI skills; Google: 100K scholarships Ghana; Intel/AfDB: 3M trained	Huawei: 120K Africans trained; FOCAC China-Africa digital center
Strategy	AI Action Plan: export full stack to allies; US-Africa Tech Collaboration; BUILD Act vs BRI	Digital BRI; low prices; telecom installed base; “small & beautiful” FOCAC 2024

6quat.4 Impact of US Protectionism on Africa: Five Transmission Channels

US AI protectionism affects the African continent through five structural channels.

First channel: restricted access to frontier compute. The Tier 2 classification imposes quantitative caps on advanced GPU exports. For a continent where the compute deficit is already the most acute in the world, these restrictions exacerbate the gap. The 12,000 GPUs of the Cassava/Nvidia project represent a modest volume compared to the millions of GPUs deployed by US hyperscalers. Access to frontier model training (GPT-5, Claude, Gemini) remains dependent on US cloud, creating structural dependency.

Second channel: technological bifurcation. African countries unable to access US compute are turning to Chinese solutions. This bifurcation is already visible: Kenyan startups are using DeepSeek and Qwen, Ethiopia is deploying Huawei, FOCAC structures China-Africa digital cooperation. Ultimately, two incompatible AI ecosystems could coexist on the continent, with implications for interoperability, standards, and data governance.¹⁹

Third channel: amplified brain drain. Africa represents only 3% of the global AI talent pool. The absence of local compute pushes the most qualified researchers toward the US, Europe, or the Emirates. US protectionism, by concentrating frontier compute on US soil, reinforces the gravitational pull of the American hub for African talent. The Deep Learning Indaba program in South Africa and Masakhane initiatives attempt to counter this trend, but the disparity of means is considerable.²⁰

Fourth channel: asymmetric energy risk. Developing AI data centers requires reliable and abundant electricity. Yet 600 million Africans still lack access to electricity. Data center operators like Raxio (Uganda, Ethiopia, Mozambique, Côte d'Ivoire, DRC, Angola) must invest in their own power lines and maintain backup diesel generators (90,000 liters stored for a 1.5 MW data center).²¹ Kenya's geothermal and Morocco's renewable advantages are exceptions, not the rule.

Fifth channel: regulatory deficit. Only 16 of 54 African countries had launched national AI strategies by July 2025. The African Union's Continental AI Strategy (2025–2030) structures a common framework in two phases, but the absence of harmonized regulation creates a vacuum that external actors (US and China) exploit to impose their standards.²² The contrast with the European AI Act is striking: where Europe imposes conditions, Africa endures them.

6quat.5 Strategic Assets and Windows of Opportunity

Despite these constraints, Africa possesses specific advantages that US AI protectionism could, paradoxically, enhance.

Selective energy advantage. Kenya (geothermal), Morocco (solar and wind), South Africa (nuclear and renewables), and Côte d’Ivoire (solar: 37.5 MWp at Boundiali, target 45% renewable by 2030) offer competitive energy corridors for data centers. The power of energy as a “moat” for data centers is now recognized: McKinsey notes that power availability, not demand density, increasingly determines siting decisions.²³

Geographic node position. Morocco (15 km from Europe, EMEA hub) and Egypt (Europe-Africa-Middle East crossroads) can become AI compute export nodes, serving European and Middle Eastern workloads at lower energy cost. This nearshore positioning is comparable to India’s for IT services, transposed to AI infrastructure.

Linguistic diversity as NLP opportunity. With over 1,000 languages spoken, Africa represents a unique innovation ground for natural language processing models. The Masakhane (NLP for African languages) and UlizaLlama projects develop specific models for Kinyarwanda, Swahili, Yoruba, Xhosa, and Zulu. Microsoft has committed \$5.5 million to the LINGUA Africa program.²⁴

Fintech market and mobile data. Africa’s fintech ecosystem (M-Pesa, OPay, Flutterwave) generates massive transactional data volumes, exploitable for training AI-specific models (credit scoring, fraud detection, financial inclusion). Africa will have 2,400+ AI-specialized companies, 41% of which are startups, and the continental AI market is projected to grow from \$4.5 billion in 2025 to \$16.5 billion by 2030 (27% annual growth).²⁵

6quat.6 Africa-Specific Scenarios

Applying the 2×2 scenario matrix (Chapter V) to the African continent generates four trajectories.

	Passive African Response	Active African Response
Moderate US Protectionism	S1 — Dependent Stagnation Limited US compute access; passive imported cloud consumption; continuous brain drain; AI ~1% of African GDP 2030	S2 — Targeted Catch-Up Operational AI Factories; Kenya/Morocco energy corridors; African language NLP; AI ~3% of GDP 2030
Intense US Protectionism	S3 — Imposed Bifurcation Shift to Chinese ecosystem; Huawei/DeepSeek dependency; standards fragmentation; surveillance risk	S4 — Digital Non-Alignment AU continental strategy; US/China/open-source multi-sourcing; data sovereignty; local open models

Scenario S4 (“digital non-alignment”) would constitute the most adapted African response, combining technological multi-sourcing (Nvidia/Huawei/open-source),

data sovereignty, and development of specific open models (African languages, agriculture, health). However, its realization requires continental regulatory coherence (AU Strategy) and massive investment estimated at over \$7 billion to bridge deficits in data, compute, and skills. Scenario S3 (imposed bifurcation) is the most likely in the absence of coordinated action, reproducing for AI the telecom dependency already established by Huawei.²⁶

6quat.7 Synthesis: The Continent of the Last Digital Chance

Africa concentrates the sharpest paradoxes of the global technological reshaping. The most compute-deficient continent (< 1% of global capacity), yet endowed with the greatest demographic potential (1.4 billion inhabitants, median age ~19) and strategic energy resources (geothermal, solar, hydro). The market most coveted by both blocs (US and China), yet the least regulatorily equipped to negotiate the terms of this competition.

Table 22 — CACI Synthesis and African Position

Indicator	Africa (2025)	US (2025)	Ratio
DC Capacity (GW IT)	0.78–1.17	53.7	×46–69
DC Investment (\$B)	~2	660–690	×330–345
AI GPUs (Nvidia)	12,000 (3–4yr target)	Millions	×100+
AI Talent	3% of global pool	~40% of global pool	×13
AI Market (\$B)	4.5 (2025) → 16.5 (2030)	~200 (2025)	×44 (2025)

US AI protectionism acts on Africa as a *digital inequality amplifier*. By limiting access to frontier compute for Tier 2 countries (virtually all of Africa) while concentrating compute, talent, and innovation on US soil, it deepens an already abyssal gap. Simultaneously, it opens a strategic boulevard for China, which possesses both the installed base (Huawei telecom) and the alternative offering (Ascend, DeepSeek) to capture this market.

For Africa, the challenge is not to choose between the United States and China, but to **build negotiating capacity** vis-à-vis both. This requires four conditions: (i) accelerating the AU Continental AI Strategy (Phase I 2025–2026), (ii) leveraging competitive energy corridors (Kenyan geothermal, Moroccan solar), (iii) investing in specific open models (African language NLP, AI for agriculture and health), and (iv) collectively negotiating compute access conditions with US *and* Chinese suppliers, rather than enduring the competition.

The window of opportunity is similar to the one identified for Europe (2026–2028) but the stakes differ: it is not about catching up on a lost advantage, but about preventing a structural lag from becoming permanent lock-in. As the New Lines Institute summarizes, the Cassava/Nvidia collaboration “is less about technology than about determining whether Africa will be a producer or merely a consumer in the global AI economy.”²⁷

Notes

- ¹ WEF (December 2025), “Investment in green computing can unlock \$1.5t in Africa.” Africa Data Centres Association: Africa represents 18% of global population but less than 1% of global DC capacity.
- ² African Energy Chamber (December 2025), “Data Centers Could Be the Spark Africa’s Power Sector Needs.” 223 DCs across 38 countries. Mordor Intelligence (January 2026): Africa DC market valued at \$1.94B in 2025.
- ³ WEF (December 2025), *ibid.* 7 million GPU hours of unmet demand; only 5% of AI innovators have access to advanced compute.
- ⁴ McKinsey (November 2025), “Building data centers for Africa’s unique market dynamics.” Projection 1.5–2.2 GW by 2030. Mordor Intelligence (2026): CAGR 14.46% Africa DC market.
- ⁵ Cassava Technologies / Bloomberg (April 2025): Nvidia GPU cost estimated at \$45,000–\$60,000 per unit. Nvidia controls ~93% of global GPU market.
- ⁶ Ecofin Agency (February 2026), “Huawei Expands Chinese AI-Integrated Data Centre Solutions to Ethiopian Banking Sector.” Huawei first to deploy operational AI in sub-Saharan Africa. Ainvest/GSMA: Huawei dominates 70% of African 4G backbone.
- ⁷ Microsoft (March 2025), ZAR 5.4B announcement through 2027. AWS (2024): \$1.7B Cape region. Mordor Intelligence: SA AI DC market from \$70M (2025) to \$572M (2031), CAGR 42%.
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- ¹² Carnegie Endowment (September 2025), “Understanding Africa’s AI Governance Landscape.” Cassava Technologies: 12,000 GPUs across 5 countries (SA, Egypt, Nigeria, Kenya, Morocco).
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- ¹⁴ Bloomberg (April 2025), “Nvidia, Cassava’s AI Factory in Africa Tie-Up to Cost \$720 Million.” TechCabal (October 2025): Nvidia equity investment in Cassava. New Lines Institute (July 2025): largest private tech infrastructure commitment in Africa.
- ¹⁵ Semafor (July 2025), “Google raises African AI bet”: \$37M specifically for AI in Africa, within \$1B global commitment. Microsoft (February 2026): 124 million people connected in Africa out of 299 million globally.
- ¹⁶ Rest of World (November 2025), “Huawei pushes AI and cloud into emerging markets after U.S. ban.” Huawei present since the 1990s. Forrester (November 2025): Ascend 910C one generation behind most advanced US offerings.
- ¹⁷ Africa Defense Forum (November 2025), “China’s DeepSeek Chatbot Expansion in Africa Raises Alarms.” DeepSeek 94% cheaper than ChatGPT. Data stored on servers accessible to Chinese government. Ainvest (August 2025): Huawei trained 120,000 Africans.
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- ¹⁹ Rest of World (November 2025), *ibid.* Rebecca Arcesati, Mercator Institute: “We are seeing a bifurcation of the global AI stack.”
- ²⁰ African Union (May 2025), “Africa Declares AI a Strategic Priority.” 83% of AI startup funding in Q1 2025 to Kenya, Nigeria, South Africa, Egypt. Only 3% of global AI talent in Africa.
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- ²² Intelpoint (August 2025): 16 of 54 African countries with national AI strategy. White & Case: AU Continental Strategy Phase I (2025–2026) focusing on governance + resource mobilization.
- ²³ McKinsey (November 2025), *ibid.* “Power availability, not demand density, increasingly determines siting decisions.” African Energy Chamber (2025): Kenya 60%+ renewable; Côte d’Ivoire 37.5 MWp solar Boundiali.
- ²⁴ Microsoft LINGUA Africa: \$5.5M for AI models for underrepresented African languages. Mastercard (August 2025), “AI in Africa”: over 1,000 languages spoken, Masakhane and UlizaLlama projects.
- ²⁵ Tech In Africa (October 2025), “North vs. Sub-Saharan Africa: AI Investment Trends.” African AI market from \$4.51B (2025) to \$16.53B (2030), CAGR 27.42%. 2,400+ AI companies, 41% startups.

²⁶ Carnegie Endowment (2025), *ibid.* Estimated \$7B+ needed to bridge data, compute, and skills deficits. AU Continental AI Strategy: Phase II (2028) for core project implementation.

²⁷ New Lines Institute (July 2025), “Accelerating U.S.-Africa Tech Collaboration.” Analysis of Cassava/Nvidia partnership as model for US-Africa technological engagement.